

How much of a nervous-system model does behaviour identify?

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Abstract

Whole-organism simulators reproduce behaviour from thousands of parameters; which of them behaviour determines is unknown. Perturbing each of a nematode model’s 3076 connectome weights, with observables outnumbering parameters, behaviour determines eight directions at 90% of the curvature, 47 at 99%. A *Drosophila* optic-lobe model agrees at 14 of 330: in two phyla, behaviour leaves the great majority of a connectome-scale space free. Chemical synapses carry 87% of the curvature and gap junctions 13%; the conductances left free are the fast voltage-gated calcium currents. Cell models with known truth verify it: small neuronal models recover exactly, MAPK matches behaviour with wrong parameters. Further aspects of one action add little, their subspaces overlapping more than they complement, whereas an action with different functional demands reaches directions the first cannot. What a fit to behaviour pins down is measurable, set by the diversity of behaviours elicited and separable from what it leaves free.

Introduction

A central premise of mechanistic whole-organism simulators is that fitting them to behaviour recovers the biophysics that produces it. Decades of work on conductance-based models, however, from the stomatogastric ganglion (1, 2) to the broader theory of sloppy models (3, 4), establish that many parameter sets produce indistinguishable dynamics: the map from parameters to behaviour is many-to-one (5–7). Whether, and how strongly, this degeneracy holds at the scale of a *whole-organism* nervous system, with thousands of synaptic, junctional and channel parameters driving a behaving body, has not been measured. The consequence is direct: if behaviour constrains only a low-dimensional subspace, calibration recovers that subspace and not the remainder, and what it leaves unconstrained are candidate directions along which biology can vary without behavioural consequence.

Here we measure the behavioural-equivalence manifold of four published whole-organism motor simulators (BAAIWorm (8), modWorm (9), flybody (10), and NeuroMechFly v2 / FlyGym (11, 12)), which are built on physics engines for musculoskeletal control (13). To test generality, we apply the same measurement to a connectome-constrained *Drosophila* visual circuit, the canonical crustacean pyloric population, and a mammalian controller; and to verify it against ground truth, to eleven cell models whose parameters are known exactly, spanning neuronal excitability and intracellular signalling. In each we measure the manifold of behaviourally equivalent parameter sets, identify the constrained subspace, and ask whether its geometry recurs across systems and in biology.

The components required to do this at scale are now available. Connectome-constrained whole-organism simulators rest on increasingly complete wiring diagrams (14–16) on community simulation frameworks (17–19), and on the whole-brain neural recordings they aim to reproduce (20–22);

high-throughput tracking reduces behaviour to low-dimensional kinematic descriptors (23–25); and parameter inference is increasingly amortised by simulation-based methods (26–28) or performed by black-box optimisers (29, 30); behaviour-based parameter generation for whole-organism simulators has recently been demonstrated from tracking video alone (31), and causal-model estimation for the fly brain argues that its dominant dynamics are carried by a small fraction of the wiring (32). What none of them settles is *how much* of a whole-organism parameter set behaviour determines: the identifiability question (33, 34) that sloppy-model theory answers for small systems (3, 4, 35) and pyloric degeneracy for a single circuit (1, 2), and a prerequisite for behaviourally grounded digital organisms (36, 37).

Results

The measurement needs a system whose forward map from biophysics to behaviour is fully specified and cheap enough to probe densely. Published whole-organism simulators qualify: every synaptic, junctional and channel parameter is explicit, and behaviour can be re-simulated under controlled perturbation. The primary system is the nematode, for which two independently constructed models exist on the same connectome, so any geometry we measure can be checked against a second implementation of the same animal. Around this core we assemble a comparison set spanning four orders of parameter count, probed by one estimator (Fig. 1).

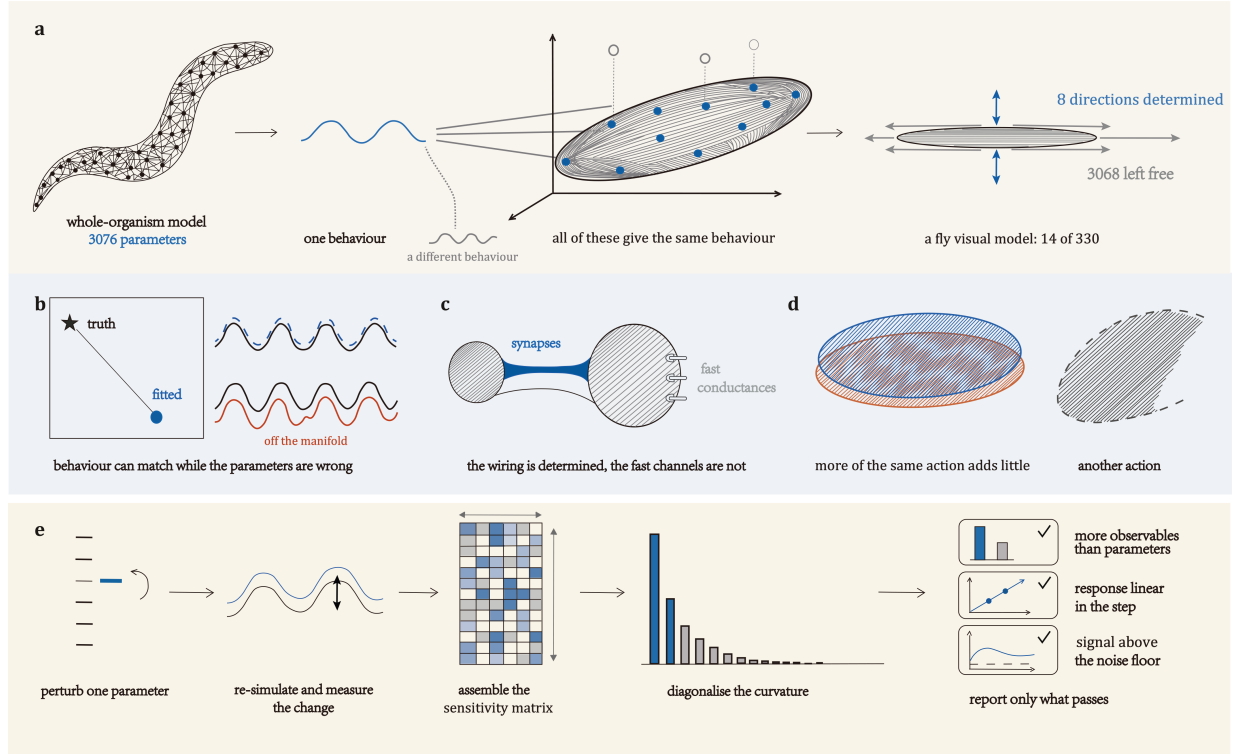


Figure 1: What behaviour determines about a nervous-system model, and how we measure it. **a**, A whole-organism simulator maps the 3076 chemical-synapse and gap-junction weights of the nematode connectome onto behaviour, and that map is many-to-one: the parameter sets producing one and the same behaviour form a behavioural-equivalence manifold, which is extremely anisotropic. Behaviour determines 8 directions at 90% of the curvature and leaves 3068 free; a *Drosophila* optic-lobe model gives 14 of 330, so in two phyla behaviour leaves the great majority of a connectome-scale space free. **b**, Where the true parameters are known, at the cell scale, a fit can match behaviour exactly while the parameters remain far from the truth, so a perfect behavioural fit does not imply recovered parameters, while a displacement of the same size taken off the manifold changes the behaviour outright; where behaviour does identify the parameters, recovery is exact. **c**, What is determined and what is free is biophysically specific: chemical synapses carry 87% of the curvature and gap junctions 13%, while the fast voltage-gated calcium conductances are left free. **d**, The identified set depends on which behaviour is elicited. Further aspects of one action re-measure combinations that action already determines (principal-angle cosine 0.54 against a random-subspace null of 0.31, $p=0.016$), whereas an action with a different functional demand determines directions the first cannot reach: chemotaxis and locomotion each render four directions of the circuit distinguishable and the two together six. **e**, The measurement. Each parameter is perturbed in turn, behaviour is re-simulated and the change in the observables recorded, the sensitivity Jacobian is assembled with observables as its rows and parameters as its columns, and its Gauss–Newton curvature $H = J^T W J$ is diagonalised. A dimension is reported only if the observables outnumber the parameters, the response is proportional to the perturbation across the step sizes used, and the signal exceeds the simulator’s own realisation noise; dimensions failing any of the three are withdrawn.

Behaviour converges while parameters remain dispersed

How much of a connectome-scale model can behaviour reach? On the BAAIWorm nematode model (136 cells; the 3076 connectome weights) we perturbed every weight in turn and read the motor-neuron membrane potentials, 80 cells over time, as 9600 observables, so observables outnumber parameters and the curvature rank is not capped by the measurement. The behavioural Gauss–Newton curvature has effective dimension 8 at 90% and 47 at 99% of 3076, participation ratio 2.55: of three thousand connection directions, behaviour constrains eight. Only 4 weights produce no measurable change, so the flat directions are degeneracy, not unconnected parameters (Fig. 2a). The measurement is differentiable and noise-free: the response is proportional to the step to four significant figures over a twentyfold range, and repeat simulations are bit-identical (Supplementary §1.5). The number describes a region and not a point, and a circuit and not a readout. At four displaced operating points the dimension stays at 8–12 and the stiff subspaces coincide at cosine 0.89 against a random null of 0.15; read through the model’s own motor output, the 96 muscle activations, the same connections give 3 / 11, so the layer closest to behaviour is the most concentrated of the two (Supplementary §5.3).

The behavioural-equivalence manifold is low-dimensional

The same geometry appears at every parameter scale, provided the observable count does not limit it. A Gauss–Newton Hessian from n_{obs} observables has rank at most n_{obs} , so a low effective dimension read through a few summary statistics says as much about the measurement as about the model. On the 48 FlyGym actuator gains, ten summary scalars give 1, whereas 4200 joint angles give 11 (90%) / 31 (99%) against a noise floor of 7 (Supplementary §5.1); on modWorm the answer does not move between six scalars and 7,200 posture observables. Across parameter counts of 4, 12, 48 and 3076, behaviour is set by a low-dimensional stiff subspace embedded in a high-dimensional flat manifold (Fig. 4b). Two choices could in principle manufacture that number, and both are measured. Sweeping the spectral-mass threshold from 80% to 99% moves the modWorm count from 2 to 4 of its five live mechanisms and the BAAIWorm class-gain count from 1 to 3 of 20, with the threshold-free participation ratios, 2.23 and 1.34, inside those ranges. Scrambling each behaviour’s Jacobian while preserving its marginals raises the dimension from 2.5–3.0 to 4.0 and the participation ratio from 1.8 to 3.5–3.9, and the estimator recovers constructed spectra of known rank exactly (Supplementary §6.4, §6.8, §6.9).

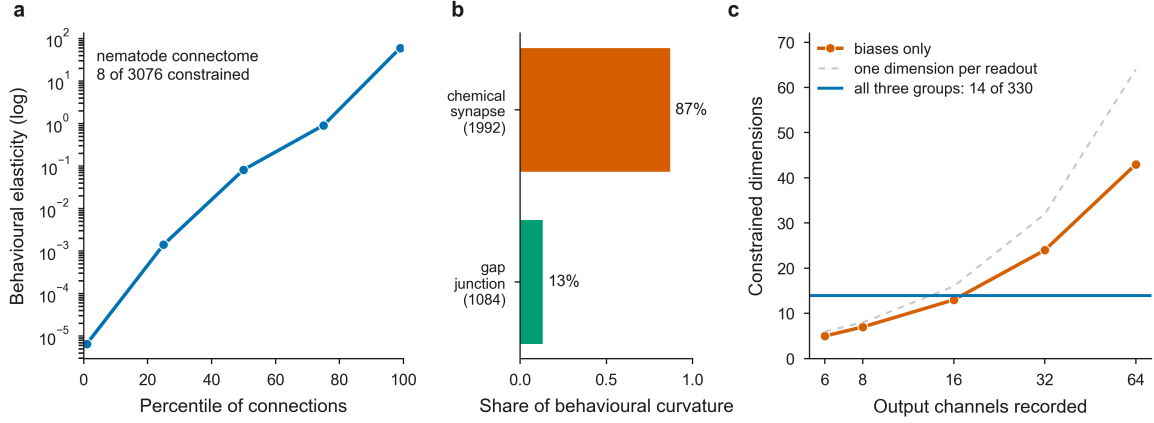


Figure 2: What behaviour constrains in two connectome-scale models, and what sets the number. **a** Behavioural sensitivity of the 3076 synaptic and gap-junction weights of the BAAIWorm nematode connectome, perturbed individually and read through 9600 observables, the membrane potentials of 80 motor neurons over time, so that observables outnumber parameters. Points are the stored percentiles of the per-connection elasticity, which span seven orders of magnitude. Effective dimension 8 (90%) / 47 (99%), participation ratio 2.55: 3068 directions are flat, and only 4 weights have no measurable effect at all. Read instead through the model’s own motor output, the 96 body-wall muscle activations, the same connections give 3 / 11. **b** The same curvature split by connection type. Chemical synapses carry 87% of it from 1992 connections and gap junctions 13% from 1084, so per connection a chemical synapse contributes 3.6 times more, while the median elasticities differ by $1.20\times$: the split lives in the heavy tail. **c** The flyvis *Drosophila* optic-lobe model. Vermillion: effective dimension of the 65 per-cell-type biases as the number of recorded output channels is reduced from 64 to 6; grey dashed: one constrained dimension per readout. Blue line: the model measured over all three of its free parameter groups, 14 of 330 sampled parameters. A nematode motor circuit and a fly visual circuit thus agree that behaviour reaches a few per cent of a connectome-scale parameter space, and the bias-only figure of 43 of 65 tracks the readout count rather than the circuit: an effective dimension measured through fewer observables than parameters is a statement about the measurement. Both models pass a differentiability check (the response is proportional to the perturbation over a twentyfold range in the nematode and across two decades in the fly), and repeat nematode simulations are bit-identical, so the finite-difference noise floor there is zero. Curvature is the Gauss–Newton form $H = J^T W J$; effective dimension is the smallest number of eigenvalues reaching the stated fraction of the spectral mass.

The manifold is the neural control of eigenworm posture

The nematode behavioural manifold is the parameter-space image of the animal’s approximately four-dimensional postural repertoire. Real *C. elegans* posture from the OpenWorm Movement Database occupies 3.79–4.14 effective dimensions, four eigenworms capturing 94–96% of postural variance, reproducing the classical eigenworm decomposition (23). Two independently built worm models generate posture overlapping this basis, at mean principal-angle $\cos 0.77$ for modWorm and $\cos 0.68$ – 0.69 for BAAIWorm. The models’ repertoires are lower-dimensional than the animal’s (1.66 and 2.76–3.30), and the agreement holds for the subspace rather than for individual modes, exceeding random four-dimensional subspaces (modWorm $p=0.001$, BAAIWorm $p=0.04$ within a conservative ten-dimensional ambient; Supplementary §6.1). Stiff mechanisms displace the eigen-

worm trajectory more than sloppy ones, by a ratio of 1.17. What the effective rank tracks is the repertoire’s intrinsic dimensionality and not the number of observables recorded: replacing six summary kinematics with 7,200–14,400 posture observables leaves it at 2–3 of 7 (Supplementary §6.2). The identifiable subspace and the neural control of posture are therefore two descriptions of one low-dimensional object, matched empirically across two models and one animal (Fig. 3a).

Functionally different actions extend the identified set

What a repertoire adds separates into two questions with opposite answers. Further aspects of one action converge on the subspace that action already determines. Four distinct locomotor presets each constrained 2–4 of 7 mechanism directions at mean pairwise stiff-subspace $\cos 0.54$, against $\cos 0.31$ and a span of 3.50 dimensions (95% interval 3.00–3.88) for randomly oriented directions: more aligned than chance ($p=0.016$ for the cosine, $p=0.020$ for the span, 20,000 draws) while still spanning 2.98 of 4 (Supplementary §6.4). Their union reached 5 of 7 at 99% of the curvature and 2 at 90%; across twelve behaviours it saturated at 4 and 2, leaving 2–3 directions that no behaviour constrains, an irreducible degenerate core. The fly behaves the same way, 12.25 to 13.0 of 48 actuator directions over four commands, an increment of 0.75 against a single-behaviour spread of $sd 3.0$. Measuring one action more finely therefore approaches a limit, and that limit is the action’s own subspace.

A different action does not. In BAAIWorm, perturbing the chemosensory parameters altered behaviour $7.15\times$ more during chemotaxis (sensitivity 0.872) than during plain locomotion in a weak gradient (0.122), whereas perturbing the motor parameters showed the opposite contrast (0.109 versus 0.357; Supplementary §6.3): the axis a behaviour renders stiff is the axis its function requires. As subspaces, chemotaxis renders four directions distinguishable and locomotion four, the two combined six, so each reaches two the other cannot. That is a rank statement and needs no threshold; the two directions it adds are the smallest of the six, at 0.30% and 0.26% of the combined curvature against 1.51% and 1.03% for the third and fourth, so they enter the spectrum alongside the directions already counted rather than at its floor (Supplementary §6.10). The constrained parameters change identity accordingly: of the four neuron classes carrying most curvature the two actions share only the ventral-cord A-type motor neurons, exchanging AWC and AIZ for the inhibitory DD and VD. This matches the anatomical separation of the chemosensory navigation circuit from the ventral-cord locomotor circuit (38, 39), and models in which premotor and motor elements generate the rhythm (40, 41). Where the eigenvectors are resolved, the stiff directions are combinations of two or three mechanism classes (participation 2.88–3.25 of 16) while the sloppy ones are near-single classes (1.02), so what behaviour leaves free is individual conductances rather than a coupled combination (Supplementary §6.7; Fig. 3b,c). In this parameterisation the free directions are therefore closable one conductance at a time, which the collective account of sloppiness (3, 35) does not anticipate.

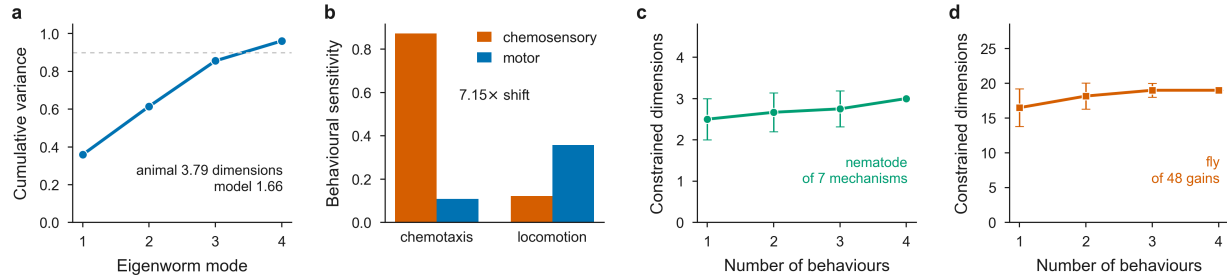


Figure 3: What behaviour identifies is the postural control space, and each action tiles a different part of it. **a** Cumulative postural variance across eigenworm modes for real *C. elegans* (OpenWorm Movement Database, $n=8432$ frames): four modes carry 96% of the variance, and the threshold-free posture dimension is 3.79 in the animal against 1.66 in the model, so the identifiable subspace is of the same order as the animal’s own postural space. **b** Behavioural sensitivity of the chemosensory and the motor parameters under food-directed chemotaxis and under plain locomotion in a weak gradient. The chemosensory parameters are stiff only during chemotaxis, a $7.15\times$ context shift, and the motor parameters show the opposite contrast: each action renders stiff the axis its function requires. **c,d** Constrained dimensions against the number of behaviours, averaged over every subset of each size so the curve does not depend on one ordering, for the nematode (7 mechanisms) and the fly (48 actuator gains). Both saturate: adding further aspects of the same action returns progressively less, and a residual set of directions is never constrained. Each system keeps its own axis because the three parameterisations do not support a comparison of absolute dimensions; the saturating shape is what is invariant.

The same geometry in a second phylum

The same organisation appears in a second phylum and in a sensory rather than a motor circuit. The flyvis model of the *Drosophila* optic lobe is built on the fly connectome and carries 734 free parameters in three groups: neuron biases, membrane time constants and synaptic strengths. Probed across all three and read through its visual output, its curvature has effective dimension 14 at 90% and 32 at 99% of 330 sampled parameters, its response proportional to the step across two decades (Supplementary §5.7). A visual circuit in one phylum and a motor circuit in another therefore return the same answer. Within the fly, the FlyGym walking controller gives 11 of 48 actuator gains read through 4200 joint-angle observables, and adding behaviours does not move it beyond the single-behaviour spread (Supplementary §6.5), so the two fly models agree with each other as well as with the worm. A mammalian controller extends the union over six behaviours, 3.7 to 12.0 of 38 actuators, and has not saturated within the set tested (Supplementary §6.5). Models parameterised differently are not comparable by absolute dimension, and what recurs across all of them is the shape: a stiff subspace of a few directions inside a flat manifold of hundreds.

Ground truth verifies the geometry where it can be known

The whole-organism models have no true parameter vector; at the cell scale the curvature can be checked against truth. Across eleven cell models measured identically and recovered from random starts independent of the truth, both ends of the geometry appear: the 2–4-parameter neuronal models come back exactly (error $< 10^{-4}$), while the canonical MAPK cascade (42), the family in which sloppiness was first described (3), matches behaviour to 10^{-3} with its parameters 40% from the truth, farther than they started. A perfect behavioural fit does not imply recovered parameters. Truth also bounds the effective dimension’s meaning: across a parameter-count $\times$ observable grid

on one neuron it is 1 in every cell while achieved recovery error spans 4–56%; it measures curvature concentration, not an upper bound on recoverable parameters (Supplementary §7). On the worm model an independent multi-start probe ranks the identifiable axes identically ($\rho=0.96$, $p=5\times 10^{-4}$), with simulation-based calibration as a control.

Ground truth also resolves an apparent disagreement with the degeneracy literature, which reports conductance-based models to be far less identifiable than the exact recovery above. The disagreement is one of protocol: measuring a single Hodgkin–Huxley neuron through twelve combinations of parameter count and assay, what moves the answer is how many conductances are searched. Three recovered against a seven-point current–frequency curve end 3.8% from the truth, eight against the identical assay 38% away, inside the degenerate range the literature describes. Nor is the small constrained fraction of the connectome-scale model a consequence of its size: adding identical cells to a fixed sparse network, from 6 to 192 parameters, moves that fraction only from 0.33 to 0.30 while the dimension itself rises at close to one direction per added cell (Supplementary §7.3, §7.4).

Stiff and sloppy directions are biophysically interpretable

Scaling each named conductance uniformly across cells and reading the full motor-neuron trajectory gives an effective dimension of 3 (90%) / 6 (99%) over sixteen conductance-class gains, with elasticities spanning three orders of magnitude. The stiffest axes are the inward rectifier IRK, the sodium leak NCA and the potassium conductances EGL-36 and EGL-2, whereas the fast voltage-gated calcium channels EGL-19 (1.77) and UNC-2 (0.17) are among the sloppiest, with SLO-2 (0.57–0.78) and SLO-1/EGL-19 (0.09); KCNL (15.1) and SLO-1/UNC-2 (6.68) sit mid-rank (Supplementary §5.2). Behaviour is thus set by how cells are connected and by their passive and slow properties, and is largely insensitive to the fast calcium conductances, documented as homeostatically compensated in small circuits (2). These identities are resolved at the level of the mechanism class. Refining one mechanism to per-cell resolution holds the dimension at 1–2 while placing 99.7% of the curvature outside the uniform direction, so the number survives the refinement and the question of which cells carry it is a separate measurement. Most of the degeneracy lies in the connectome: 3068 of 3076 directions are flat, and the surviving curvature is uneven, 1992 chemical synapses carrying 87% against 13% from 1084 gap junctions, so per connection a synapse matters 3.6 times more. The medians differ by $1.20\times$, which places the split in the heavy tail of strong connections rather than in the typical one (Supplementary §5.3). Probed identically, modWorm (coupled-ODE, 7 mechanisms) and BAAIWorm (NEURON, 20 named channels) agree that the network-connectivity axes are stiff (stiff fraction 0.83 / 0.87) and the single-cell biophysical axes sloppy (0.08 / 0.16), so “connectivity stiff, single-cell sloppy” reproduces across independent implementations of the same organism (Supplementary §6.6).

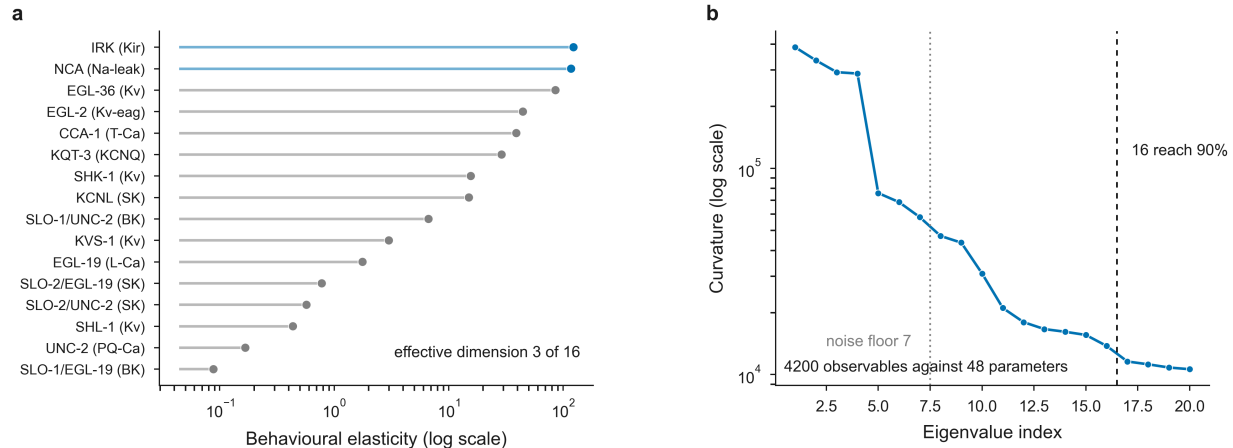


Figure 4: Stiff and sloppy structure at named-channel and higher-dimensional scale.

a Behavioural elasticity of the BAAIWorm named conductances, read through the full motor-neuron trajectory, ordered from stiff (top) to sloppy (bottom). The stiffest are the inward rectifier IRK (123.5), the sodium leak NCA (117.7) and the potassium conductances EGL-36 (86.0) and EGL-2 (44.6); the sloppiest are SHL-1 (0.44) and the fast voltage-gated calcium channel UNC-2 (0.17), with EGL-19 (1.77) close behind. The calcium-activated potassium conductances divide: KCNL (15.1) and SLO-1/UNC-2 (6.68) rank mid-table, SLO-2 (0.57–0.78) and SLO-1/EGL-19 (0.09) at the bottom. Effective dimension 3 (90%) / 6 (99%) over all sixteen conductances, none of which is silent. Behaviour is therefore set by wiring and by passive and slow properties, and is largely insensitive to the fast voltage-gated calcium currents, which are the currents documented as homeostatically compensated in small circuits. **b** Eigenspectrum of the behavioural Gauss–Newton curvature of the 48 FlyGym actuator gains, measured through 4200 joint-angle observables rather than ten summary scalars, at a perturbation of 0.25 where this controller’s response is proportional to the step. Effective dimension 11 (90%) / 31 (99%) of 48, against a noise floor of 7 / 22 obtained from seed-to-seed variation at identical parameters. Enlarging the observable set moves the figure from 1 to 11, so an effective dimension measured through a small observable set is a statement about the measurement. Curvature is the Gauss–Newton form $H = J^T W J$, computed by central finite differences and therefore positive semi-definite.

Convergent observations in independent biological data

Four datasets that use none of our forward models show the same structure at four levels of description (Fig. 5). (1) *Pyloric degeneracy*. Here the behaviourally equivalent set is enumerated rather than inferred: 2365 conductance sets produce the same triphasic rhythm (1) and span 23.2 of 31 dimensions, so fixing the rhythm leaves most of the conductance space free, structured by correlation among 1.1% of pairs. Its response saturates in the step for both observable sets, so it fails the second gate and no curvature dimension is reported for it (Supplementary §2.1). (2) *Neuronal property variability*. Across $n=1914$ Allen cortical neurons the rate and shape properties span a coefficient of variation from 0.34 to 1.85 on a manifold of participation dimension 4.78 of 11, the membrane-potential properties being excluded since that measure depends on the zero of a millivolt scale. (3) *Gene conservation*. Across 40 *C. elegans* ion-channel and synaptic genes, ortholog identity spans 77.9–99.9%, the most conserved being the gap-junction gene **unc-9**, consistent with connectivity being stiff in both worm models. (4) *Behavioural variation*. Across $n=61$ isogenic N2 worms (683 Tierpsy features), cross-individual behaviour occupies a participation dimension

of 9.75: genetically identical animals vary on a roughly ten-dimensional manifold. What biology conserves and what it permits to vary are ordered consistently with the models' identifiable and degenerate axes.

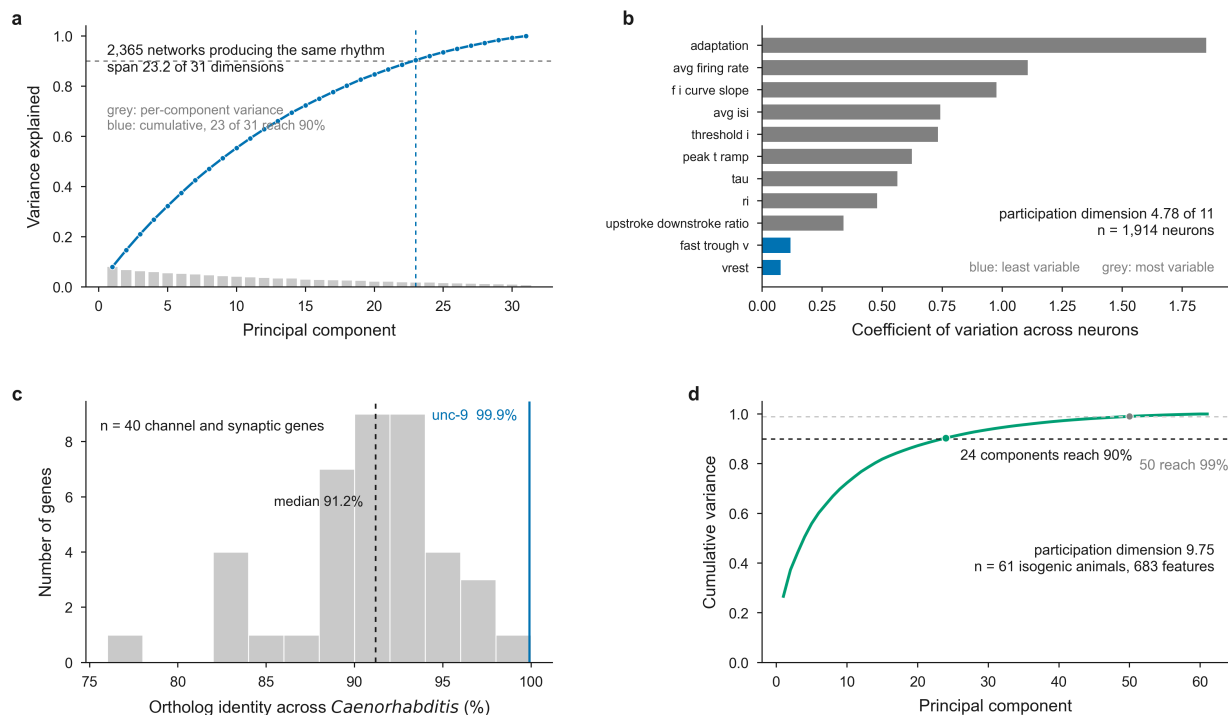


Figure 5: Low-dimensional stiff and sloppy structure in four biological datasets that use none of the simulators. **a** Prinz–Marder pyloric model population: the 2365 published conductance sets producing the canonical triphasic rhythm, shown as the variance spectrum of the parameter cloud. Grey, variance per principal component; blue, cumulative; dashed lines mark 90%. The behaviourally equivalent set spans a participation dimension of 23.2 of 31, with 23 components reaching 90%, so fixing the rhythm leaves most of the conductance space free. No curvature is shown for this system: its response saturates in the step for both observable sets, so it fails the second gate and its effective dimension is withdrawn (Supplementary §2.1). **b** Electrophysiological property variability across Allen Cell Types cortical neurons. Among the rate and shape properties the coefficient of variation spans 0.34 (upstroke–downstroke ratio) to 1.85 (adaptation); participation dimension 4.78 of 11; $n=1914$ neurons. The two membrane-potential properties, in blue, are excluded from that comparison, a coefficient of variation being undefined independently of the zero of a millivolt scale. **c** Ortholog identity of 40 *C. elegans* ion-channel and synaptic genes across *Caenorhabditis*, ranging from 77.9 to 99.9%. The gap-junction gene *unc-9* is the single most conserved at 99.9%. **d** Cross-individual behavioural manifold of isogenic N2 worms, 683 Tierpsy features, $n=61$ animals; participation dimension 9.75, with 24 principal components reaching 90% of the variance and 50 reaching 99%. All four datasets are public and no simulator is used in any of them; the comparison is made at the level of low dimensionality and direction, not of individual eigenvectors.

Discussion

We report a measurement of what behaviour can and cannot identify in whole-organism nervous-system models. In the nematode model built on the connectome, behaviour constrains eight of three thousand connection directions, and the conductances it barely sees are the fast voltage-gated calcium currents. Measuring further aspects of one action adds little: in the fly the behaviours compared are descending turn commands driving one pattern generator, in the nematode touch at two sites and two intensities, and the subspaces they determine overlap far more than they complement. Actions that engage different circuits behave otherwise: chemotaxis and locomotion render different axes stiff, four directions each and six together, the two additional directions entering at the scale of the third and fourth. Escape and feeding remain untested, for want of a simulator that drives them independently, a requirement on the models before it is a question about the animals.

The organising principle is that behaviour reports on a small, structured part of a model. Any one behaviour pins a few stiff parameter combinations, and different behaviours pin the combinations their function requires; between aspects of one action the subspaces overlap heavily and the union saturates at once.

This recasts degeneracy as a measurable resource rather than a fixed limit, and says where parameter recovery should be posed. What behaviour identifies is a few coordinates on a parameter vector rather than the vector itself; in the cell models where the truth is known, recovery along those coordinates is exact while recovery of the full vector is not. The identified set grows with the number of functionally distinct actions a simulator can be driven through, making assay diversity rather than recording duration the quantity to optimise, but it does not grow without limit: after the whole repertoire has been used a residual degenerate core remains, and those are the candidate neutral directions.

The result lifts the sloppy-models picture (3, 4, 35) and pyloric degeneracy (1, 6) from single circuits to behaving whole organisms, and explains why behaviour-based calibration recovers a working parameter set without unique biophysical values: it recovers the stiff union, not the core. That two connectome-constrained models in different phyla, one visual and one motor, return the same few per cent indicates that the number reflects how much of a nervous system behaviour transmits rather than a property of one simulator.

How much was recorded sets what can be found, and by how much is now measurable. A Gauss–Newton curvature built from n_{obs} observables has rank at most n_{obs} , and the models track that bound: reading the visual model through fewer output channels moves its effective dimension almost one for one with the channel count, and the FlyGym figure moves from 1 to 11 when ten summary scalars are replaced by 4200 joint angles. An effective dimension read through fewer observables than parameters is therefore a property of the assay, and the same holds for a step size below the proportional range or a signal below the simulator’s own noise. Every dimension here carries its observable count, step-size scan and noise floor, and the three conditions have teeth: they retire the pyloric circuit, whose response saturates in the step for every observable set we tried, and they move the fly figure from 16 to 11 once it is measured where the response is proportional.

Three limitations bound these results. Each dimension is conditional on a choice of observables, which no principle fixes uniquely. The conductance decomposition resolves mechanism classes rather than individual cells, so it names which currents behaviour leaves free and not which cells express them. And behaviour locates a perturbation only to within a set: the mechanisms it can detect are mutually confusable at cosine above 0.9, and the alignment with the biological datasets holds at the level of dimensionality and direction rather than eigenvector by eigenvector (Supplementary §5.4, §5.5).

The strongest outstanding test is prospective: predicting from the model which perturbations should produce the largest behavioural change, and testing that ordering on data not used to build it. Such a test requires perturbations resolved at the coupled parameter combinations the geometry identifies rather than at single genes or channels, and defining those endpoints in advance protects the statistics when independent samples are few.

Behavioural identifiability is therefore a quantity that can be measured rather than assumed, and its extent is set by the diversity of behaviours elicited. The stiff subspace is the part of a fitted model that the assay pins down and that can be read as biology; the free directions name what it leaves open and which further behaviours would close. The same geometry prices an assay before it is run: one that adds another aspect of the same action re-measures known combinations, one that changes what the circuit must do reaches new ones. This applies to any mechanistic simulator fitted to behaviour.

Methods

Manifold drift. For each simulator, N optimisations (separable CMA-ES or SPSA) from independent starts are fitted to one fixed behavioural target; we report behavioural recovery, the normalised per-dimension dispersion of the N recovered parameter vectors, and the PCA participation dimension of the solution cloud. **Behavioural curvature.** All curvature reported here is the Gauss–Newton form $H = J^\top W J$, built from the finite-difference observable Jacobian $J_{ik} = \partial \text{obs}_i / \partial \theta_k$ and a diagonal positive semi-definite whitening matrix W of inverse per-observable scales. H is therefore positive semi-definite by construction and equals the Fisher information of the parameters given the behavioural observables, so its eigenvalues are non-negative and its effective dimension is the local practical-identifiability rank in the standard sense (3, 33). Eigenvalues are obtained by direct eigendecomposition (low dimension) or Lanczos iteration (high dimension). The effective dimension at level q is the smallest k such that $\sum_{i \leq k} \lambda_i \geq q \sum_i \lambda_i$ with eigenvalues sorted in decreasing order; we report $q = 0.90$ and $q = 0.99$, together with the threshold-free participation ratio $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ (Supplementary §6.9). Curvature is evaluated at the published operating point, and its point dependence is measured directly: on the connectome-scale model, four displaced operating points give effective dimension 8–12 with stiff subspaces coinciding across points at cosine 0.89 against a random null of 0.15 (Supplementary §5.3); on modWorm, five displaced optima and twenty-four random points all give a low effective dimension (Supplementary §6.8). A direction counts as flat when λ falls to the finite-difference floor; residual eigenvalues of magnitude below that floor, of either sign, are numerical noise and are treated as zero. We adopt it as the primary cross-simulator metric because it is robust to the choice of observable and parameterisation, unlike a raw perturbation-sensitivity ratio, and because the manifold-drift probe provides an independent, optimisation-based cross-check that agrees with it ($\rho=0.96$). **Eigenworm identity.** Real *C. elegans* skeletons (OWMD N2) and model body postures are reduced to tangent-angle eigenmodes by PCA; we report the posture effective dimension, the top-four cumulative variance, and the principal-angle cosine between model and real eigenworm subspaces, with significance assessed against random subspaces of matched dimension. **Complementary tiling and saturation.** Per-behaviour stiff subspaces are taken as the leading Hessian eigenvectors under each behavioural target; we report pairwise stiff-subspace cosines, the union effective dimension over a behaviour set, and the saturation curve of union dimension against the number of behaviours. Context dependence (BAAIWorm) is the ratio of a parameter class’s behavioural sensitivity between two behaviours (chemotaxis versus weak-gradient locomotion). The same union/saturation construction is applied to the rodent (Virtual Rodent / dm_control). **Cell-scale ground truth.**

Eleven published cell models (five neuronal, six biochemical oscillators) are measured with the identical estimator and recovered from random starts $\theta^* e^{\mathcal{U}(-\ln f, \ln f)}$ at $f=1.5-10$, with a phase-invariant feature loss computed at full resolution; a fit counts as converged below behavioural distance 0.02 and only converged fits contribute parameter errors (Supplementary §7). **Calibration.** Per-dimension rank-based simulation-based calibration (43, 44) of an amortised neural posterior; a dimension passes when the per-axis Kolmogorov–Smirnov test is non-significant at $\alpha=0.05$. **External datasets.** The Prinz–Marder pyloric population, read as the geometry of the 2365 published conductance sets that produce the canonical rhythm rather than as a curvature, since its response saturates in the step; Allen Cell Types electrophysiology; WormBase ParaSite / Ensembl Metazoa ortholog identities; and OpenWorm Movement Database isogenic-N2 Tierpsy features. All four are public and independent of the simulators measured here. Full specifications, cohort sizes and reproduction commands are given in the Supplementary Information.

Data availability

All datasets analysed in this study are publicly available. The external biological references are: the Prinz–Marder conductance-based model population of the stomatogastric ganglion (1); the Allen Cell Types electrophysiology database (<https://celltypes.brain-map.org>); ortholog identities from WormBase ParaSite and Ensembl Metazoa (<https://parasite.wormbase.org>); and the OpenWorm Movement Database (<http://movement.openworm.org>). The whole-organism simulators are the published releases of BAAIWorm (8), modWorm (9), flybody (10), NeuroMechFly v2 / FlyGym (11), the connectome-constrained optic-lobe model of Lappalainen et al. (45), larvaworld, and the Virtual Rodent of the `dm_control` suite. All quantities reported in the main text and Supplementary Information are provided as result files, one file per analysis as named in each Supplementary section, in the deposit at <https://github.com/ChenxiHeCam/behavioural-parameter-manifold>, archived at Zenodo (<https://doi.org/10.5281/zenodo.21594739>).

Code availability

All analysis code, the per-experiment result files, and a container that reproduces every reported quantity and figure from those files are available at <https://github.com/ChenxiHeCam/behavioural-parameter-manifold> under an MIT licence, archived at Zenodo (<https://doi.org/10.5281/zenodo.21594739>). The reproduction entry point verifies each numerical claim against its source file and regenerates the figures.

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Author contributions

C.H. conceived the study, designed and performed all analyses, and wrote the manuscript.

Competing interests

The author declares no competing interests.

References

- [1] Astrid A Prinz, Dirk Bucher, and Eve Marder. Similar network activity from disparate circuit parameters. *Nature Neuroscience*, 7(12):1345–1352, 2004.
- [2] Eve Marder and Jean-Marc Goaillard. Variability, compensation and homeostasis in neuron and network function. *Nature Reviews Neuroscience*, 7:563–574, 2006.
- [3] Ryan N Gutenkunst, Joshua J Waterfall, Fergal P Casey, Kevin S Brown, Christopher R Myers, and James P Sethna. Universally sloppy parameter sensitivities in systems biology models. *PLoS Computational Biology*, 3(10):e189, 2007.
- [4] Mark K. Transtrum, Benjamin B. Machta, Kevin S. Brown, Bryan C. Daniels, Christopher R. Myers, and James P. Sethna. Perspective: Sloppiness and emergent theories in physics, biology, and beyond. *Journal of Chemical Physics*, 143:010901, 2015.
- [5] Mark S. Goldman, Jorge Golowasch, Eve Marder, and L. F. Abbott. Global structure, robustness, and modulation of neuronal models. *Journal of Neuroscience*, 21:5229–5238, 2001.
- [6] Eve Marder and Adam L Taylor. Multiple models to capture the variability in biological neurons and networks. *Nature Neuroscience*, 14(2):133–138, 2011. doi: 10.1038/nn.2735.
- [7] Gerald M Edelman and Joseph A Gally. Degeneracy and complexity in biological systems. *Proceedings of the National Academy of Sciences*, 98(24):13763–13768, 2001.
- [8] Mengdi Zhao, Ning Wang, Xinyu Jiang, Xiaohan Ma, Haoran Ma, Gan He, Kai Du, Lei Ma, and Tiejun Huang. An integrative data-driven model simulating *C. elegans* brain, body and environment interactions. *Nature Computational Science*, 4:106–120, 2024.
- [9] Eli Shlizerman et al. modworm: a modular muscular-skeletal c. elegans simulator. *Preprint*, 2025.
- [10] Roman Vaxenburg, Igor Siwanowicz, Josh Merel, Alice A Robie, Carmen Morrow, Guido Novati, Sam Stupski, Gwyneth M Card, and Srinivas C Turaga. Whole-body simulation of *drosophila* reveals motor control of behavior. *Nature*, 2025. flybody release, Figshare DOI 10.25378/janelia.25309105.
- [11] Sibow Wang-Chen et al. Neuromechfly v2: simulating embodied sensorimotor control in adult drosophila. *Nature Methods*, 2024.
- [12] Victor Lobato-Rios, Shravan Tata Ramalingasetty, Pembe Gizem Özdil, Jonathan Arreguit, Auke Jan Ijspeert, and Pavan Ramdya. NeuroMechFly, a neuromechanical model of adult *Drosophila melanogaster*. *Nature Methods*, 19(5):620–627, 2022.
- [13] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5033, 2012.

- [14] John G White, Eileen Southgate, J Nichol Thomson, and Sydney Brenner. The structure of the nervous system of the nematode *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society of London B*, 314(1165):1–340, 1986.
- [15] Steven J Cook, Travis A Jarrell, Christopher A Brittin, et al. Whole-animal connectomes of both *Caenorhabditis elegans* sexes. *Nature*, 571(7763):63–71, 2019.
- [16] Sven Dorkenwald, Arie Matsliah, Amy R. Sterling, et al. Neuronal wiring diagram of an adult brain. *Nature*, 634:124–138, 2024.
- [17] Balázs Szigeti, Pádraig Gleeson, Michael Vella, et al. OpenWorm: An open-science approach to modeling *Caenorhabditis elegans*. *Frontiers in Computational Neuroscience*, 8:137, 2014.
- [18] Pádraig Gleeson, David Lung, Radu Grober, et al. c302: a multiscale framework for modelling the nervous system of *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society B*, 373(1758):20170379, 2018.
- [19] Michael L. Hines and Nicholas T. Carnevale. The NEURON simulation environment. *Neural Computation*, 9(6):1179–1209, 1997.
- [20] Saul Kato, Harris S Kaplan, Tina Schrödel, et al. Global brain dynamics embed the motor command sequence of *Caenorhabditis elegans*. *Cell*, 163(3):656–669, 2015.
- [21] Jeffrey P. Nguyen, Frederick B. Shipley, Ashley N. Linder, George S. Plummer, Mochi Liu, Sagar U. Setru, Joshua W. Shaevitz, and Andrew M. Leifer. Whole-brain calcium imaging with cellular resolution in freely behaving *Caenorhabditis elegans*. *Proceedings of the National Academy of Sciences*, 113(8):E1074–E1081, 2016.
- [22] Francesco Randi and Andrew M Leifer. Multi-modal whole-organism neuroscience: Integrating behavior, neural dynamics, and physiology in *C. elegans*. *Current Opinion in Neurobiology*, 79:102684, 2023.
- [23] Greg J Stephens, Bethany Johnson-Kerner, William Bialek, and William S Ryu. Dimensionality and dynamics in the behavior of *C. elegans*. *PLoS Computational Biology*, 4(4):e1000028, 2008.
- [24] Avelino Javier, Michael Currie, Chee Wai Lee, Jim Hokanson, Kezhi Li, Céline N. Martineau, Eviatar Yemini, Laura J. Grundy, Chris Li, QueeLim Ch’ng, William R. Schafer, Ellen A. A. Nollen, Rex Kerr, and André E. X. Brown. An open-source platform for analyzing and sharing worm-behavior data. *Nature Methods*, 15(9):645–646, 2018.
- [25] Talmo D. Pereira, Joshua W. Shaevitz, and Mala Murthy. Quantifying behavior to understand the brain. *Nature Neuroscience*, 23(12):1537–1549, 2020.
- [26] Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, 2020.
- [27] Pedro J Gonçalves, Jan-Matthis Lueckmann, Michael Deistler, et al. Training deep neural density estimators to identify mechanistic models of neural dynamics. *eLife*, 9:e56261, 2020.
- [28] Alvaro Tejero-Cantero, Jan Boelts, Michael Deistler, Jan-Matthis Lueckmann, Conor Durkan, Pedro J Gonçalves, David S Greenberg, and Jakob H Macke. sbi: A toolkit for simulation-based inference. *Journal of Open Source Software*, 5(52):2505, 2020.

- [29] James C Spall. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE Transactions on Automatic Control*, 37(3):332–341, 1992.
- [30] Nikolaus Hansen and Andreas Ostermeier. Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation*, 9(2):159–195, 2001.
- [31] Chenxi He. Generating behaviour-equivalent parameters for whole-organism simulators from tracking video. Preprint, 2026. URL <https://doi.org/10.21203/rs.3.rs-10486658/v1>. Preprint.
- [32] Dean A. Pospisil, Max J. Aragon, Sven Dorkenwald, Arie Matsliah, Amy R. Sterling, Philipp Schlegel, Szi-chieh Yu, Claire E. McKellar, Marta Costa, Katharina Eichler, Gregory S. X. E. Jefferis, Mala Murthy, and Jonathan W. Pillow. The fly connectome reveals a path to the effectome. *Nature*, 634:201–209, 2024. doi: 10.1038/s41586-024-07982-0.
- [33] Alejandro F. Villaverde and Julio R. Banga. Reverse engineering and identification in systems biology: strategies, perspectives and challenges. *Journal of the Royal Society Interface*, 11(91): 20130505, 2014.
- [34] Andrea Saltelli, Marco Ratto, Terry Andres, Francesca Campolongo, Jessica Cariboni, Debora Gatelli, Michaela Saisana, and Stefano Tarantola. *Global Sensitivity Analysis: The Primer*. John Wiley & Sons, 2008.
- [35] James P. Sethna. Sloppy models, parameter uncertainty, and the role of experimental design. *Molecular BioSystems*, 6:1846, 2010.
- [36] Jonathan R. Karr, Jayodita C. Sanghvi, Derek N. Macklin, Miriam V. Gutschow, Jared M. Jacobs, Benjamin Bolival Jr, Nacyra Assad-Garcia, John I. Glass, and Markus W. Covert. A whole-cell computational model predicts phenotype from genotype. *Cell*, 150(2):389–401, 2012.
- [37] Anthony Zador, Sean Escola, Blake Richards, Bence Ölveczky, Yoshua Bengio, Kwabena Boahen, et al. Catalyzing next-generation artificial intelligence through NeuroAI. *Nature Communications*, 14:1597, 2023.
- [38] Jesse M. Gray, Joseph J. Hill, and Cornelia I. Bargmann. A circuit for navigation in *Caenorhabditis elegans*. *PNAS*, 102:3184–3191, 2005.
- [39] Cornelia I Bargmann. Beyond the connectome: How neuromodulators shape neural circuits. *BioEssays*, 34(6):458–465, 2012.
- [40] Quan Wen, Michelle D. Po, Elizabeth Hulme, Sway Chen, Xinyu Liu, Sze Wai Kwok, et al. Proprioceptive coupling within motor neurons drives *C. elegans* forward locomotion. *Neuron*, 76(4):750–761, 2012.
- [41] Eduardo J Izquierdo and Randall D Beer. Connecting a connectome to behavior: an ensemble of neuroanatomical models of *C. elegans* klinotaxis. *PLoS Computational Biology*, 14(2): e1005968, 2018.
- [42] Boris N. Kholodenko. Negative feedback and ultrasensitivity can bring about oscillations in the mitogen-activated protein kinase cascades. *European Journal of Biochemistry*, 267(6): 1583–1588, 2000. doi: 10.1046/j.1432-1327.2000.01197.x.

- [43] Sean Talts, Michael Betancourt, Daniel Simpson, Aki Vehtari, and Andrew Gelman. Validating Bayesian inference algorithms with simulation-based calibration. *arXiv preprint arXiv:1804.06788*, 2018.
- [44] Joeri Hermans, Arnaud Delaunoy, François Rozet, Antoine Wehenkel, Volodimir Begy, and Gilles Louppe. A trust crisis in simulation-based inference? Your posterior approximations can be unfaithful. In *Transactions on Machine Learning Research*, 2022.
- [45] Janne K. Lappalainen, Fabian D. Tschopp, Sridhama Prakhya, Mason McGill, Aljoscha Nern, Kazunori Shinomiya, Shin-ya Takemura, Eyal Gruntman, Jakob H. Macke, and Srinivas C. Turaga. Connectome-constrained networks predict neural activity across the fly visual system. *Nature*, 634:1132–1140, 2024.